Glymmy: Unified AI Glycemic Companion & Smartwatch Interface

Proposal Track: Chronic Disease Management & Multimodal AI | **Team:** Glymmy | **Date:** November 2026

Executive Hook: *Over 537 million people manage diabetes globally, yet their critical health data is scattered across disconnected silos: CGM apps, smartwatch fitness trackers, unread lab PDFs, and manual diaries.* ***Glymmy*** *is an AI-powered unified health companion that fuses Continuous Glucose Monitoring, wearable biometrics (HR, sleep, steps), and computer vision food logging into local predictive intelligence (Gemma 4). Its signature frontend innovation is the* ***SF-LG (Space-Filling Line Graph) Greedy Space Allocation Engine****, delivering non-occluding, actionable clinical intelligence directly onto circular smartwatch displays.*

1. The Problem: Data Fragmentation & Clinical Inefficiency

Managing Type 1 and Type 2 diabetes requires continuous synthesis of high-frequency biological signals. However, patients and clinicians face three compounding bottlenecks:

Siloed Data Domain	Current Patient Experience	Clinical Limitation / Unmet Need
Continuous Glucose (CGM)	Dexcom or Libre app shows numbers and simple alerts.	Isolated from meal composition, heart rate, and insulin timing.
Wearable Biometrics	Apple Watch/Garmin tracks HR, sleep, and steps.	Data is siloed as generic fitness, ignoring glycemic impact and nocturnal hypoglycemia.
Food & Nutrition	Manual logging in calorie counters or photo albums.	High friction, no automated carbohydrate estimation, no predictive spike guidance.
Diagnostic Labs	Paper reports or portal PDFs every 3–6 months.	Patients cannot interpret biomarkers (HbA1c, eGFR, lipid panels) between appointments.

2. The Glymmy Solution & Technical Architecture

Glymmy unifies raw multimodal inputs into a private, local-first intelligence pipeline. Rather than inventing proprietary hardware, Glymmy leverages devices patients already own.

System Layer	Technologies Used	Functional Capability
Multimodal Ingestion	BLE GATT, HealthKit, Nightscout/xDrip+ API	Streams 5-min glucose, 10-min vitals (HR, SpO2, sleep, steps) and photo uploads.
AI & Predictive Core	Gemma 4 (Local via Ollama), ChromaDB RAG	Multimodal vision for meal carb extraction, post-prandial spike forecasting, and clinical QA.
Backend Platform	FastAPI, Python 3.12, Pydantic v2, uv	Sub-50ms API endpoints, asynchronous SSE alert streaming, Dubosson data normalization.
Clinical Data Sharing	HL7 FHIR R4 Bundle + QR Generator	Encodes longitudinal glucose profile into a 10-second QR scan for doctor visits.

Frontend Clients	Next.js 16 Web Dashboard + Vite Smartwatch	Comprehensive clinical web portal and standalone circular smartwatch interactive UI.

3. Core Innovation: SF-LG Smartwatch Space Allocation Engine

A critical bottleneck on modern smartwatches is viewport occlusion: on a 300×300px circular bezel, displaying actionable advice typically forces the line graph to shrink or covers the data. Glymmy implements the **Space-Filling Line Graph (SF-LG) Greedy Space Allocation Algorithm**:

Algorithm Operation:
• **Coordinate Projection:** Maps glucose readings (40–240 mg/dL) onto bounded circular SVG coordinates.
• **Valleys Discovery:** Scans all X-axis slices (width ≥ 38px) and computes local curve extrema (Ymin, Ymax).
• **Candidate Rectangles:** Identifies vacant 'headroom' above the curve and 'footroom' below the curve.
• **Greedy Area Allocation:** Sorts candidates by area descending and non-overlappingly assigns the top two pockets to host dynamic contextual cards.

Interactive Watch Scenario	Allocated Valley & Trigger	Clinical Impact & User Interaction
S4: Post-Meal Spike & Walk Advisor *(Signature Feature)*	Valley Above Curve Triggered by Meal Pin (52g carbs)	Shows predicted peak (192 mg/dL). User clicks 'Simulate 15-min Walk' — the CGM curve dynamically blunts to 138 mg/dL (-25 mg/dL) in real time.
S1: Point Vitals Correlation	Largest Valley Triggered by Point Tap	Reveals glucose value, trend vector (■/■/→), plus correlated Heart Rate (bpm) and active Insulin on Board (IOB).
S2: Rapid Drop & Rule of 15	Second Largest Valley Ambient Pulsing Alert	Expands to clinical rescue: 'Rule of 15: Eat 15g fast carbs & retest in 15m' with tachycardia monitoring.
S3: Comparative Analysis	Two Largest Valleys Tap 2 Points	Displays Point A vs Point B with exact glycemic delta (Δ mg/dL) and time elapsed.
Ambient Bezel Complications	Top Apex Arc (Y: 8px to 28px)	Centered complication stack: Time (10:09) flanked by Time-in-Range (TIR 84%) and IOB (1.4U), cleanly contoured inside the circular bezel.

4. Clinical Grounding, Datasets & Validation

Glymmy is calibrated against validated, real-world clinical datasets rather than arbitrary heuristics:

Dataset / Cohort	Data Contents	Application in Glymmy Pipeline
GlucoBench (Dubosson)	9 T1D patients with continuous CGM, insulin bolus/basal, HR, and accelerometer.	Calibrates post-meal spike kinetics, exercise dips, and physiological insulin modeling.
GlucoBench (Hall et al.)	57 individuals (healthy, prediabetic, diabetic) over longitudinal periods.	Establishes glucotype subclassification and personalized food response variations.
MIMIC-IV Clinical Demo	100 intensive care patients, 107,000 lab events with reference intervals.	Powers the RAG diagnostic interpreter for blood work (lipid profiles, metabolic panels).

5. Live Demonstration Plan (3-Minute Presentation)

All functional components are operational locally without cloud or API dependencies:

Timing	Stage & Live Demo Endpoint	Key Action & Narrative Flow
0:00 – 0:30	Problem & Ramesh Persona	Introduce Ramesh (45, T2D, Kathmandu) managing 4 fragmented apps. Hook the judges.
0:30 – 1:15	Food Scanner & Gemma 4 localhost:3000	Photograph meal (dal, rice, roti). AI predicts +38 mg/dL spike in 60m and recommends a 15-min walk.
1:15 – 2:05	**SF-LG Smartwatch Demo localhost:3001**	Demonstrate greedy space allocation on the circular screen. Click 'Simulate 15-min Walk' — watch curve flatten in real time.
2:05 – 2:35	Clinical Protocols (S1 & S2) localhost:3001	Showcase correlated Heart Rate + IOB on point tap; expand 'Rule of 15' rescue card for rapid drops.
2:35 – 3:00	FHIR QR Share & Conclusion localhost:8000/docs	Generate 10-second FHIR QR code for doctor appointments. Zero new hardware. One unified companion.

6. Implementation Roadmap & Feasibility

Phase	Milestones Achieved / Targeted	Target Horizon
Hackathon MVP	• Working SF-LG Smartwatch engine with dynamic spike blunting • FastAPI backend with 9,900+ seeded records • Next.js dashboard & Gemma 4 RAG pipeline	**Completed**
Clinical V1	• Direct BLE GATT integration for Dexcom G6/G7 and FreeStyle Libre • HealthKit/WearOS on-device native bridge • PaddleOCR blood report batch importer	Months 1 – 3
Scale & Systems	• Multi-language localization (Nepali, Hindi, Spanish) • EHR integration via SMART on FHIR • Prospective clinical trial measuring TIR improvement	Months 4 – 12